

# Data Flow & Trust Boundary Map

Document Reference: DOC-03  
System: Case Ace v2.0 Client Consultation Recording & Note Drafting System  
Organisation: Citizens Advice Wandsworth (CAW)  
Status: Verified Architectural Baseline  
Classification: Official-Sensitive / Governance Pack

## 1. Visual System Architecture & Trust Boundary Map

[!NOTE]  
High-Resolution Vector SVG & Interactive Explorer:  
For high-resolution visual presentations, governance reviews, and interactive inspection:  
Standalone Vector SVG Diagram: [evidence/data-flow-diagram.svg](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/evidence/data-flow-diagram.svg)  
Interactive Architecture Explorer: [evidence/data\_flow\_architecture\_interactive.html](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/evidence/data\_flow\_architecture\_interactive.html)

### Visual Block Architecture Diagram



|                                     |               |  |                                                      |
|-------------------------------------|---------------|--|------------------------------------------------------|
| Organisations, Locations)           |               |  |                                                      |
| +-----+                             |               |  | +-----+                                              |
| +-----+                             |               |  |                                                      |
| +-----+                             |               |  |                                                      |
|                                     |               |  |                                                      |
| v                                   |               |  |                                                      |
| +-----+                             |               |  | +-----+                                              |
| (window.crypto.subtle) [C5]         |               |  | 4. Surrogate Tokenisation & Ephemeral Keyring        |
| [POSTCODE_1]                        |               |  | 'Jane Doe' -> [CLIENT_NAME_1]   'SW11 2LN' ->        |
| RAM closure*                        |               |  | *Lookup dictionary retained strictly in volatile     |
| +-----+                             |               |  | +-----+                                              |
|                                     |               |  |                                                      |
| v                                   |               |  |                                                      |
| +-----+                             |               |  | +-----+                                              |
| Screen)                             |               |  | 5. Phase 9 Redaction Review Gate (Adviser UI         |
| adds missed PII spans*              |               |  | *Adviser verifies highlighted tokens & manually      |
| +-----+                             |               |  | +-----+                                              |
|                                     |               |  |                                                      |
| v                                   |               |  |                                                      |
| +-----+                             |               |  | +-----+                                              |
| WAV) [C2/C8]                        |               |  | 6. Phase 10 Acoustic Muting & Redaction (LINEAR16    |
| audio wiped with 0s*                |               |  | *Mutes audio timestamps matching PII spans   Raw     |
| +-----+                             |               |  | +-----+                                              |
|                                     |               |  |                                                      |
| Only)                               | (Tokens Only) |  | (Muted WAV                                           |
|                                     |               |  |                                                      |
| v                                   | v             |  |                                                      |
| +-----+                             |               |  | +-----+                                              |
| +-----+                             |               |  | +-----+                                              |
| TRUST BOUNDARY 2: CAW BACKEND PROXY |               |  | 7. In-Browser Detokenisation & Adviser Professional  |
| Review [C6]                         |               |  | *Swaps [CLIENT_NAME_1] back to real client name      |
| (Google Cloud Run in europe-west2)  |               |  |                                                      |
| locally in browser RAM*             |               |  | *Adviser checks advice accuracy, edits gaps, and     |
|                                     |               |  |                                                      |
| clicks 'Sign Off'*                  |               |  |                                                      |
| +-----+                             |               |  | +-----+                                              |
| +-----+                             |               |  | +-----+                                              |
| 1. Scoped GCP STS Broker            |               |  |                                                      |
|                                     |               |  |                                                      |
| (Exchanges Entra JWT for            |               |  |                                                      |
| v                                   |               |  |                                                      |
| 15-min downscoped token)            |               |  |                                                      |
| +-----+                             |               |  | +-----+                                              |
| +-----+                             |               |  | +-----+                                              |
| +-----+                             |               |  | 8. destroySession() Deterministic Memory Destruction |
| Engine [C7]                         |               |  | Uint8Array.fill(0) wipes all audio   Drops           |
| 2. Strict Audit Log Store           |               |  |                                                      |
| WebCrypto keys   Clears RAM         |               |  |                                                      |
| (0-PII Whitelist Schema)            |               |  |                                                      |
| +-----+                             |               |  | +-----+                                              |
| +-----+                             |               |  | +-----+                                              |
| +-----+                             |               |  | +-----+                                              |
| Note)                               |               |  | (Adviser Copies Signed                               |
| v                                   |               |  | v                                                    |
| +-----+                             |               |  | +-----+                                              |
| +-----+                             |               |  | +-----+                                              |

|                                                        |                                     |
|--------------------------------------------------------|-------------------------------------|
| TRUST BOUNDARY 3: GOOGLE CLOUD (europe-west2 London)   | TRUST BOUNDARY 4: NATIONAL CASEBOOK |
| CRM                                                    |                                     |
| [Pinned to London   Zero Data Retention (ZDR) Terms]   | (Official Citizens Advice Case      |
| Management System)                                     |                                     |
|                                                        |                                     |
| 1. Google Cloud Speech-to-Text v2 (Chirp 2)            |                                     |
| +-----+                                                |                                     |
| *Receives ONLY Verified Redacted Audio*                | Casebook Web                        |
| Application                                            |                                     |
| *PII audio spans are muted silence*                    | - Adviser manually pastes finalized |
| note                                                   | - Permanent client record           |
|                                                        |                                     |
| creation                                               | - Subject to national data          |
| 2. Google Cloud Vertex AI (Gemini 1.5 Enterprise)      |                                     |
| retention rules                                        |                                     |
| *Receives ONLY Surrogate-Tokenised Text*               |                                     |
| +-----+                                                |                                     |
| *Zero prompt logging   Zero model retraining*          |                                     |
| +-----+                                                |                                     |
| +-----+                                                |                                     |
| ^                                                      |                                     |
| (SSO Authentication ONLY)                              |                                     |
| +-----+                                                |                                     |
| MICROSOFT ENTRA ID (Organisational Identity Perimeter) |                                     |
| [STRICTLY ISOLATED TO INITIAL LOGIN SCREEN]            |                                     |
| - Adviser OIDC SSO & MFA Login                         |                                     |
| - ZERO access to case notes, transcripts, or audio     |                                     |
| +-----+                                                |                                     |

## 2. Detailed Trust Boundary Analysis & Transits

### 3. Data Residue & Destruction Boundary Guarantee

When any session termination event occurs (Explicit End, Logout, Idle Timeout, Consent Withdrawal, Tab Close, Unrecoverable Error), the `destroySession()` engine executes across all internal memory partitions:

Audio Memory: All `Float32Array` and `Uint8Array` audio buffers (raw and redacted) are overwritten in-place with zeroes (`.fill(0)`).

Token Store: The token map dictionary and identifier arrays are deleted and unreferenced.

Draft & Transcript Store: All text representations of the consultation are cleared.

Web Workers: The `SessionRecoveryWorker` instance is explicitly sent a termination message and terminated.

Credentials: Outstanding short-lived GCP STS credentials are revoked and discarded.

Clipboard: If the adviser copied detokenised case note content, the system wipes the clipboard buffer (where platform permissions allow).